Span and linear combinations.

Let V be a vect. space over K. Let $S \subseteq V$ be a subset (not necess. a subspace).

Q. 1. What is the smallest linear subspace of V that contains S?

2. What do we get if we consider the subset of V obtained by all linear combinations of elements from S?

Lemma. Let V be a vect. space over K. Let $\{W_i\}_{i \in I}$ be a collection of subspaces $W_i \subseteq V$ of V, indexed by $i \in I$ (I is a set of indices).

Then $W := \bigcap_{i \in I} W_i$ is a linear subspace of V.

Proof. $0 \in W$, b.c. $0 \in W_i \ \forall i \in I$.

Let $a_1, a_2 \in K$, $v_1, v_2 \in W$.

We want to show $a_1 v_1 + a_2 v_2 \in W$. (*)

Let $j \in I$. Since $v_1, v_2 \in \bigcap_{i \in I} W_i$ we have

$v_1, v_2 \in W_j$. $\Rightarrow$ $a_1 v_1 + a_2 v_2 \in W_j$ b.c. $W_j \subseteq V$

is a subspace.

We have proved that: $\forall\, j \in I$ $a_1 v_1 + a_2 v_2 \in W_j$.

$$\Rightarrow \quad a_1 v_1 + a_2 v_2 \in \bigcap_{i \in I} W_i = W.$$

which is what we wated to prove in (*).

$\blacksquare$

<u>Definition of span.</u> Let $S \subseteq V$ be a subset.

We define $Sp(S) := \bigcap_{W \in \mathcal{N}} W$,

where $\mathcal{N} := \{ W \mid W \text{ is a subspace of } V \text{ and } S \subseteq W \}$,

i.e. the collection of all subspaces $W \subseteq V$ that contain S.

$Sp(S)$ is called the span of S (in V).

<u>Lemma.</u> (1) $Sp(S) \subseteq V$ is a linear subspace.

(2) If $W \subseteq V$ is a linear subspace and $S \subseteq W$, then $Sp(S) \subseteq W$.

In other words, $Sp(S)$ is a linear subspace of V, and moreover among those subspaces that contain S, $Sp(S)$ is the smallest one.

<u>Proof.</u> Follows quickly from the defs and from the previous lemma. Check the details!

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